MODULATION AND DEMODULATION SIMULATOR

Interactive Analog & Digital Communication Signal Simulation System

Department of Electronics And Communication (Advanced Communication Technology)
CSMSS Chh. Shahu College of Engineering, Chh. Sambhajinagar

TEAM MEMBERS & ROLL NUMBERS:

1. Gargi Deshpande - AC2148 (Team Leader)
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Dept: Electronics & Communication (ACT)

PROJECT GUIDE & STATUS:

Guide: Prof. Amruta Kadli Mam

Department: Electronics & Communication (ACT)

Current Progress: 80% COMPLETED

Presentation Deadline: 31st October 2026

Project Abstract & Simulation Scope

Modulation & Demodulation Simulation for Communication Engineering

1. System Overview

Understanding communication waveforms without expensive laboratory hardware is challenging. This simulator provides real-time mathematical generation, noise addition, and recovery of modulated signals across both time-domain and frequency-domain representations.

2. Supported Modulation Techniques

- Amplitude Modulation (AM): DSB-FC, DSB-SC, SSB-SC, and Envelope Detection.
- Frequency Modulation (FM): Narrowband & Wideband FM, Carson’s Rule, Foster-Seeley.
- Phase Modulation (PM) & Phase Demodulation.
- Digital Shift Keying: ASK, FSK, BPSK, and QPSK Constellations.
- AWGN Channel Simulation: Additive White Gaussian Noise with variable SNR.

Interactive Laboratory Utility:

Enables engineering students and faculty to tune carrier frequency, modulating frequency, modulation index, and noise levels with live FFT spectrum and audio demodulation output.

Faculty Guide: Prof. Amruta Kadli Mam | Progress: 80% Completed

Mathematical Equations & Demodulation Algorithms

Implemented Signal Generation and Filter Transfer Functions

Amplitude Modulation (AM)

s_AM(t) = Ac [1 + m * cos(2*pi*fm*t)] * cos(2*pi*fc*t)

Modulation index: m = Am / Ac

Demodulation: Diode Envelope Detector with

RC time constant: 1/fc << RC << 1/fm

Frequency Modulation (FM)

s_FM(t) = Ac * cos[2*pi*fc*t + beta * sin(2*pi*fm*t)]

Modulation index: beta = delta_f / fm

Carson's Rule Bandwidth: BW = 2 * (delta_f + fm)

Demodulation: Phase Locked Loop (PLL) & Quadrature

Fast Fourier Transform (FFT)

X(k) = sum_{n=0}^{N-1} x(n) * e^{-j*2*pi*k*n/N}

Live spectral computation showing carrier

and upper/lower sidebands in real-time.

Dynamic bandwidth monitoring & distortion check.

Digital Modulation (BPSK/QPSK)

s_BPSK(t) = sqrt(2*Eb/Tb) * b(t) * cos(2*pi*fc*t)

Coherent Costas loop demodulation.

Bit error rate (BER) vs SNR curve generation.

Constellation diagram I/Q plotting.

Researchers: Gargi Deshpande, Mayuri Jagtap, Sanika Bhalerao, Asawari Kavre | CSMSS Chh. Shahu COE

Project Milestones & Development Status

Current Progress: 80% Completed | Presentation Due: 31st October 2026

X **Milestone 1: Mathematical Waveform Equations Formulation**

Status: 100% Completed - AM, FM, and PM mathematical models coded.

X **Milestone 2: Time-Domain Waveform & Live FFT Spectrum Visualizer**

Status: 100% Completed - Canvas & Chart.js live rendering engine ready.

X **Milestone 3: Demodulation Filters & Noise Injection Module**

Status: 100% Completed - Envelope detector, PLL, and AWGN noise added.

X **Milestone 4: Interactive Web UI & Parameter Control Sliders**

Status: 100% Completed - Reactive slider tuning for frequency and modulation index.

-> **Milestone 5: Final Defense, Presentation & Faculty Evaluation**

Status: In Progress (Due 31st October 2026) before Guide Prof. Amruta Kadli Mam.

OVERALL PROGRESS: 80% COMPLETED

Verified by Prof. Amruta Kadli Mam